

Similar solids and the volume scale factor

Cambridge insert: one length ratio, three consequences — used on real models and real costs.

LESSON 40

1 × 40 MIN

GEOMETRY 11, §18 (EXTENSION)

IGCSE E21.1–21.2

LESSON CLOCK

8'

14'

12'

8'

3'

The three rules	8 min
Working backwards	14 min
Applications	12 min
Why size matters	8 min
Homework	3 min

LEARNING OBJECTIVES

- Apply the k , k^2 and k^3 rules to similar solids.
- Recover the scale factor from an area or a volume ratio.
- Solve problems on models, packaging and costs.
- Explain why large animals are proportionally thicker.

TEXTBOOK

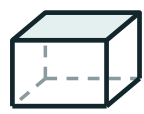
National	pp. 201–204
Cambridge	Core & Extended, pp. 226–234
Workbook	IGCSE Exercise 21.2

01

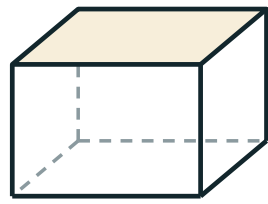
Explanation

The three rules

$$\frac{L_2}{L_1} = k \cdot \frac{S_2}{S_1} = k^2 \cdot \frac{V_2}{V_1} = k^3$$



k = 1



k = 2

lengths ×k · areas ×k² · volumes ×k³

Double every length: four times the surface, eight times the volume.

Two solids are similar when one is an enlargement of the other. All cubes are similar; all spheres are similar; two cones are similar exactly when $\frac{r}{h}$ is the same for both.

Working backwards

$$k = \sqrt{\frac{S_2}{S_1}} \quad k = \sqrt[3]{\frac{V_2}{V_1}}$$

GIVEN	K	OTHER FACTORS
volumes $8 : 27$	$\frac{2}{3}$	areas $4 : 9$
areas $16 : 25$	$\frac{4}{5}$	volumes $64 : 125$
lengths $1 : 50$	$\frac{1}{50}$	areas $1 : 2500$, volumes $1 : 125\,000$

Applications

QUESTION	WHICH FACTOR
how much paint?	k^2
how much liquid?	k^3
how much wire round the edge?	k
how much does it weigh?	k^3
how much fabric to cover it?	k^2

Example. A model at $1 : 20$ needs 0.15 litres of paint. The real object needs $0.15 \times 400 = 60$ litres — not 3.

PAINT IS AN AREA, NOT A VOLUME

A coat of paint is a thin layer over a surface, so it scales with k^2 . Filling a tank scales with k^3 . Reading which is asked is the whole question.

Why size matters

As a solid grows, its volume outruns its surface:

$$\frac{S}{V} \text{ scales as } \frac{k^2}{k^3} = \frac{1}{k}$$

THREE CONSEQUENCES

A large animal has proportionally **less** skin per kilogram, so it loses heat more slowly — which is why elephants have big ears and mice do not.

Its mass grows as k^3 but its bone cross-section only as k^2 , so it needs proportionally thicker legs.

Small objects cool, dry and dissolve faster — the reason for chopping food and grinding ore.

Doubling an animal's height multiplies its weight by 8 and its bone strength by only 4.

The scaled-up animal would break its own legs — a real constraint, first noticed by Galileo.

02 Worked examples

EXAMPLE 1 Two similar cones have volumes 54 and 128 cm³. Find the ratio of their surface areas.

$$\textcircled{1} \quad k^3 = \frac{128}{54} = \frac{64}{27}$$

$$\textcircled{2} \quad k = \frac{4}{3}$$

$$\textcircled{3} \quad k^2 = \frac{16}{9}$$

ANSWER

9 : 16

EXAMPLE 2 A model car is $1 : 24$. It needs 12 cm^2 of paint. The real car needs?

① $k = 24$, so $k^2 = 576$.

② $12 \times 576 = 6912 \text{ cm}^2$.

③ $= 0.69 \text{ m}^2$.

ANSWER

$\approx 0.69 \text{ m}^2$

EXAMPLE 3 Two similar jugs hold 0.5 and 1.5 litres. Their heights are in what ratio?

① $k^3 = 3$

② $k = \sqrt[3]{3} \approx 1.442$

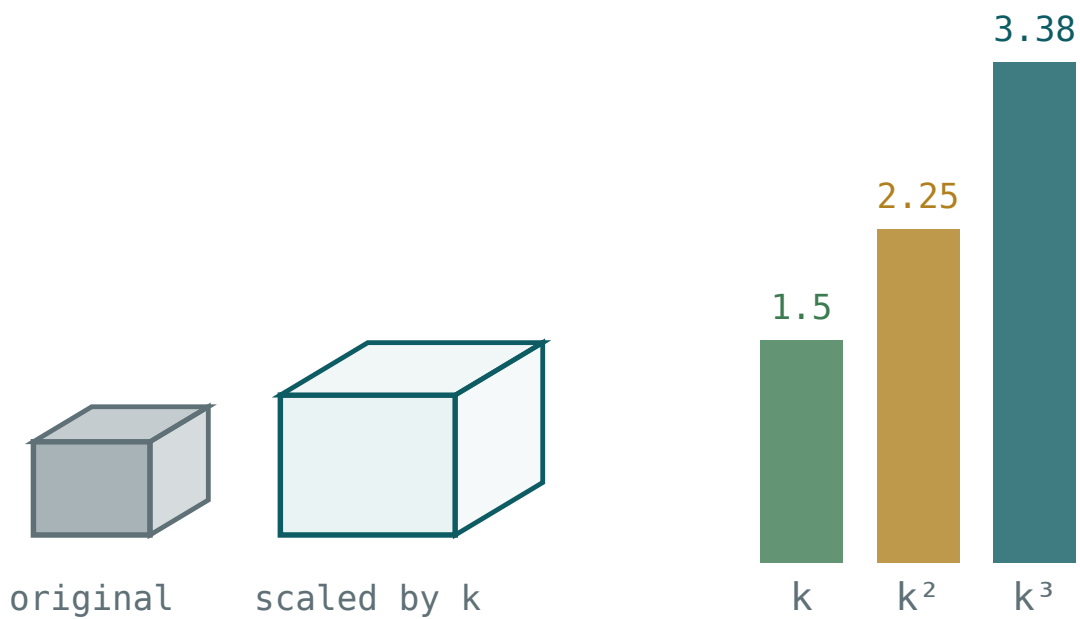
ANSWER

$1 : 1.44$

03 Interactive model

Compare two similar containers and predict, then measure, the capacity ratio.

Scaling a solid



k 1.5

length $\times$ 1.5area $\times k^2$ 2.25volume $\times k^3$ 3.38

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Similar solids	O'xshash jismlar	Подобные тела
Scale factor	O'xshashlik koeffitsienti	Коэффициент подобия
Area factor	Yuza koeffitsienti	Коэффициент площади
Volume factor	Hajm koeffitsienti	Коэффициент объёма
Model	Model	Модель
Corresponding lengths	Mos uzunliklar	Соответственные длины
Cube root	Kub ildiz	Кубический корень
Surface-area-to-volume ratio	Sirt va hajm nisbati	Отношение поверхности к объёму

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Doubling all lengths multiplies the volume by:

And the surface area by:

Volumes $8 : 27$ gives lengths:

Paint scales with:

S V scales as:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $k = 3$: area factor

ANSWER 9

2. $k = 3$: volume factor

ANSWER 27

3. $k = 0.5$: volume factor

ANSWER 0.125

4. Volumes 27 : 64: lengths

ANSWER 3 : 4

5. Areas 9 : 25: lengths

ANSWER 3 : 5

6. Paint scales with

ANSWER k^2

7. Liquid scales with

ANSWER k^3

Medium · the standard question

7 PROBLEMS

1. Volumes 54 and 128: the area ratio

ANSWER 9 : 16

2. Model 1 : 24 needing 12 cm² of paint: the real area

ANSWER ≈ 0.69 m²

3. Jugs of 0.5 and 1.5 litres: the height ratio

ANSWER 1 : 1.44

4. Model 1 : 50: the volume ratio

ANSWER 1 : 125 000

5. A statue 1 : 6 needs 0.5 litres of paint. The real one

ANSWER 18 litres

6. Two similar tins, heights 8 and 12; smaller holds 320 ml

ANSWER 1080 ml

7. Areas 49 : 81: the volume ratio

ANSWER 343 : 729

Hard · stretch and reason

7 PROBLEMS

1. Two similar solids differ in volume by 111 with $k = \frac{4}{3}$

ANSWER 81 and 192

2. A cone cut halfway up: the volume ratio of the parts

ANSWER 1 : 7

3. A 1 : 200 model bridge with deck area 80 cm²: the real area

ANSWER 320 m²

4. Why can a mouse survive a fall an elephant cannot?

ANSWER Its surface-to-mass ratio is far larger

5. A doubled animal: weight and bone strength factors

ANSWER 8 and 4

6. Two similar containers hold 2 and 6.75 litres. The height ratio

ANSWER 2 : 3

7. A sugar cube dissolves faster when crushed. Why?

ANSWER Crushing multiplies the surface area at the same volume

07 Homework

Homework — 4 tasks

State k explicitly before using k^2 or k^3 .

- Two similar cones have volumes 40 and 135 cm³. Find the ratio of their heights and of their surface areas.
- A model aeroplane is 1 : 72. Its wing area is 60 cm². Find the real wing area in m².
- Two similar bottles hold 330 and 1000 ml. Find the ratio of their heights.

4. Explain, using k^2 and k^3 , why a doubled animal could not stand.